Week 2 Insight Report – Ops Sensor Log Analysis
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CONTEXT

The raw sensor log contained numerous quality issues: missing readings (especially in Flow Rate), inconsistent zone naming, physically impossible values (e.g., Pressure > 9999 PSI, Temperature = 1500°C), and duplicate records. These problems masked true operational patterns and could lead to incorrect decisions.

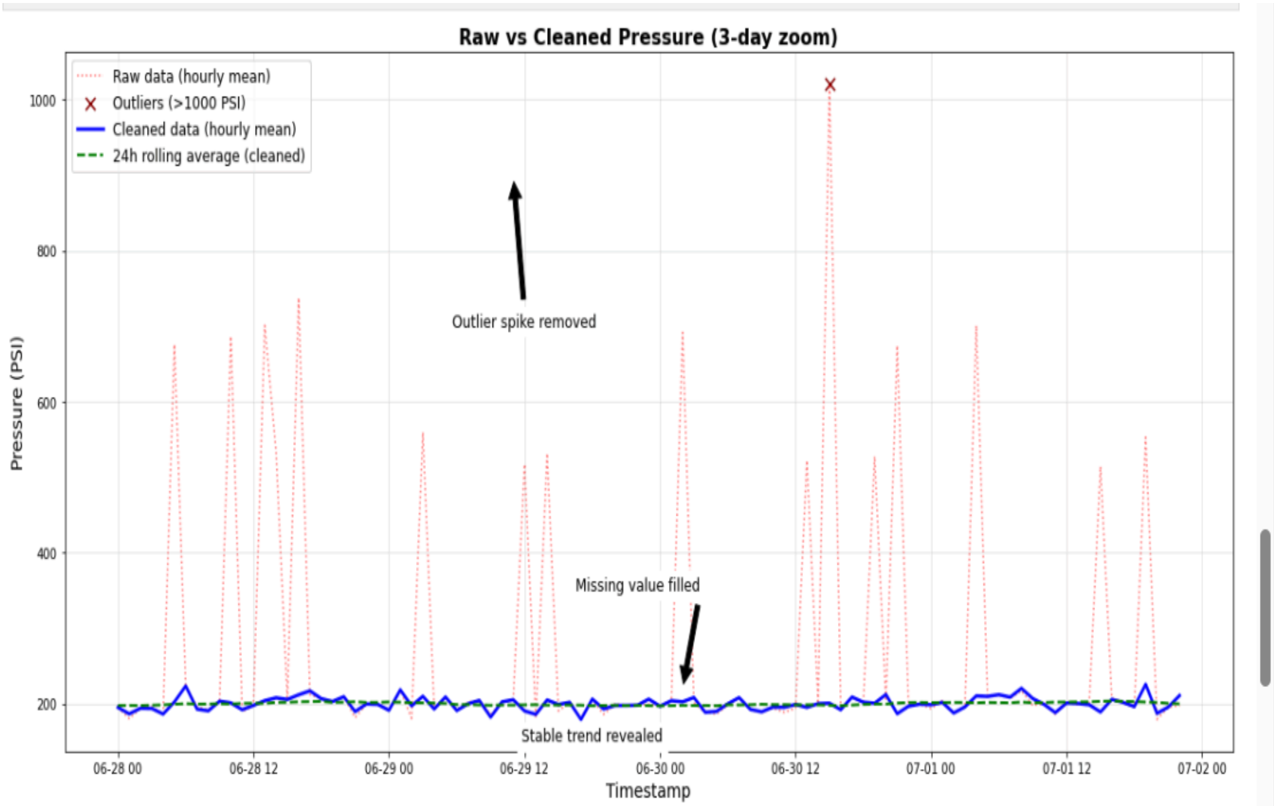

Figure 1: Pressure trend over a 3-day window. The raw data (red dotted line) contains extreme outliers (marked with X) and missing gaps. After cleaning (blue solid line), the 24-hour rolling average (green dashed line) reveals a clear daily cycle dropping at night and peaking in the afternoon

INSIGHT

After cleaning, a clear daily pressure cycle emerged: pressures drop during the Night shift and peak in the Afternoon. The raw data, with its outliers and gaps, suggested a stable process with occasional spikes. The cleaned 24-hour rolling average reveals a predictable rhythm that aligns with production schedules, something previously hidden.

ACTION

Adjust the afternoon shift's compressor setpoint to reduce peak pressure stress. The Night shift can operate at lower pressure, saving energy. Implement automated alerting when pressure deviates more than 10% from the rolling average to catch equipment issues early. I recommend a pilot test on Zone South starting next week.